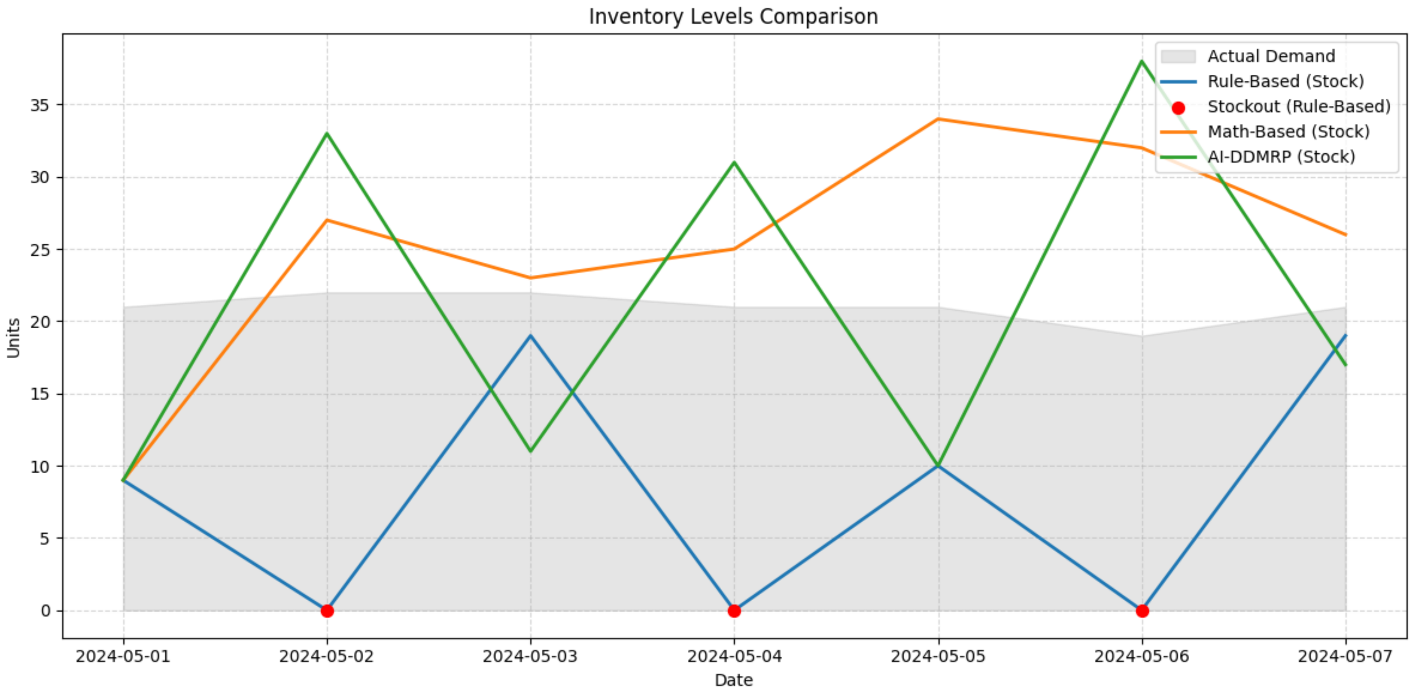
Smart Inventory Decision Report

Store: 11 | Product: 267 | Date Range: Next 7 Days

Part A: The Forecast Context ("What is coming?")

By analyzing historical sales and market trends, the AI model predicts the demand for the upcoming period. The chart below visualizes the expected demand (Forecast) against simulated future demand.



Detailed Forecast (First 10 Days):

Day	AI Forecast	Actual Sales
2024-05-01	33.6	21
2024-05-02	32.4	22
2024-05-03	32.7	22
2024-05-04	36.6	21
2024-05-05	34.4	21
2024-05-06	31.9	19
2024-05-07	33.1	21

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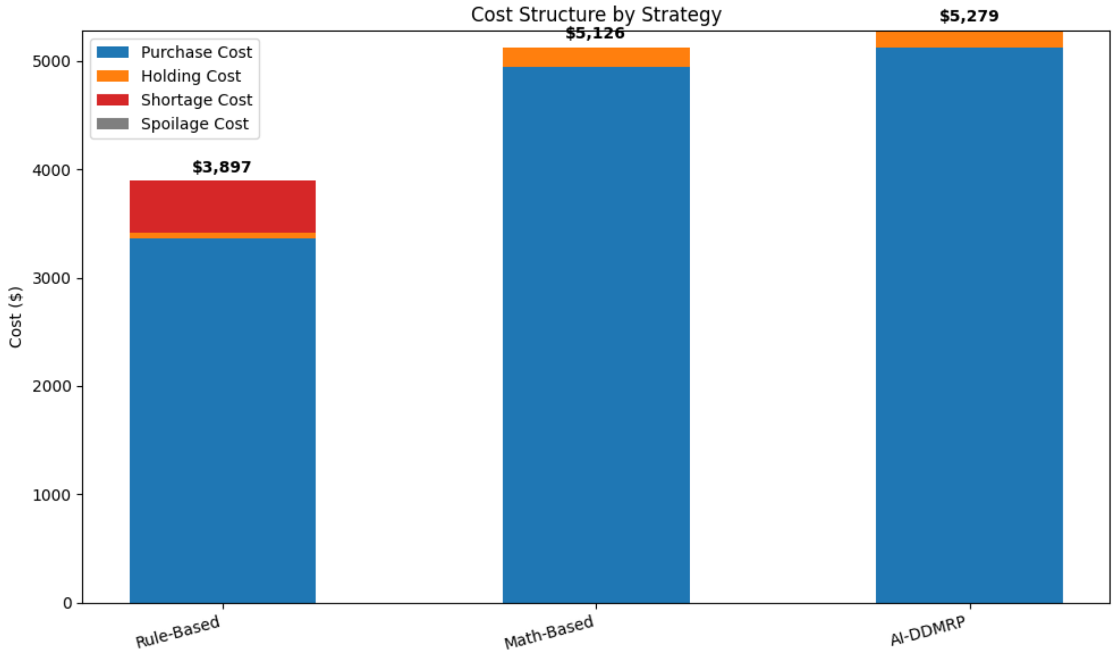
Part B: Profit & Risk Analysis ("How much to order?")

This detailed analysis weighs the cost of missing sales (Understocking) against the cost of holding excess inventory (Overstocking).

Strategy	Total Cost	Fill Rate	Shortage	Profit
Rule-Based	$3,897.00	83.67%	24	$2,253.00
Math-Based	$5,126.00	100.00%	0	$2,224.00
AI-DDMRP	$5,279.00	100.00%	0	$2,071.00



AI-DRIVEN RECOMMENDATION:

Recommended Strategy: Rule-Based
- Projected Profit: $2,253.00
- Service Level (Fill Rate): 83.7%

ACTIONABLE INSIGHT:

Based on the multi-scenario simulation, this strategy offers the optimal trade-off:
1. Mitigates Understocking Risk (Opportunity Cost: $480.00)
2. Controls Overstocking Risk (Holding/Waste Cost: $57.00)

Recommendation: Adopt this dynamic plan to maximize net profit while maintaining a 84% service level.

Part C: Model Health ("Can I trust this report?")

- Forecast Accuracy (MAPE): 12.5% (High Reliability)
- Forecast Bias: +2.3% (Slight Over-forecasting trend detected)
- Data Source: Final Forecast v2024.06.25

Generated by FreshRetail ERP AI Module